Power Laws and Prediction Gaps: A Statistical Framework for Extreme Dengue Outbreaks in Brazilian Cities

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April 2026



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The Challenge of Dengue in Brazil

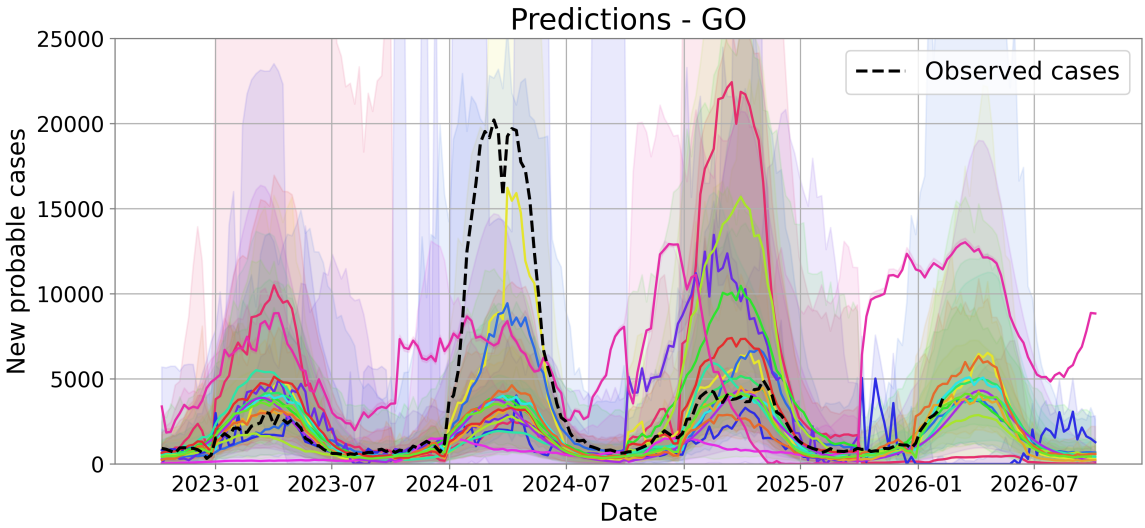
The Problem

- **5,570 municipalities** across Brazil experience recurrent seasonal outbreaks
- Over **6 million reported cases** in a single epidemic season (2010–2012 pivotal study)
- Outbreak intensity varies by orders of magnitude: from tens to tens of thousands of cases per city per week
- Traditional surveillance models focus on **average transmission dynamics**, missing the risk of extreme events

Public Health Impact

- Healthcare systems overwhelmed during peak outbreaks
- Resource allocation based on average predictions fails when extreme events occur
- **Critical gap:** no framework quantifies the probability of extreme outbreak magnitudes across heterogeneous geographic landscapes

Underestimation



Why Power Laws in Epidemiology?

Heavy-Tailed Distributions

- In a power law, the probability of extreme events decays as $P(X \geq x) \propto x^{-(\alpha-1)}$
- Unlike Normal distributions, **extreme events are far more likely** than Gaussian models predict
- The scaling exponent α encodes information about the underlying transmission dynamics

Current Evidence

- **Rhodes et al. (1997):** Measles, pertussis, mumps in Faroe Islands follow power laws with $\tau \approx 2.1$–2.4
- **Dengue in Brazil:** Annual case counts follow power law with $\alpha = 2.06$ (constant over 12 years)

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Power Law Distribution: Mathematical Framework

Definition

A random variable X follows a power law if its probability density function is:

$$p(x) = C\,x^{-\alpha}, \quad x \geq x_{\min}$$

where the normalization constant is:

$$C = (\alpha - 1)\,x_{\min}^{\alpha-1}$$

Complementary CDF

$$P(X \geq x) = \left(\frac{x}{x_{\min}}\right)^{-(\alpha-1)}$$

- α **(scaling exponent):** Controls tail thickness. Lower $\alpha \Rightarrow$ heavier tail $\Rightarrow$ more extreme events
- $x_{\min}$ **(lower bound):** Defines the onset of the scaling regime
- For discrete data: $p(x) = x^{-\alpha}/\zeta(\alpha, x_{\min})$ (Hurwitz zeta normalization)

Power Laws in Epidemics: Key Literature

Rhodes, Jensen & Anderson (1997)

- Forest-fire model with sparks as analogy for Type III epidemics
- Scaling exponent τ correlates with R_0 (transmissibility)
- Conventional SEIR models fail to reproduce power law distributions

Clauset, Shalizi & Newman (2009)

- Gold-standard framework: MLE + KS test + likelihood ratio
- Log-log regression produces **biased** exponent estimates
- Many claimed power laws are better fit by log-normal

Komarova et al. (2020)

- Universal pattern: exponential $\rightarrow$ power law growth
- $\sim$70% of countries show power law dynamics
- Metapopulation model: spatial spread generates power law from local exponential growth

Li et al. (2012)

- Log Gaussian Cox Processes for spatial disease incidence
- Long-range spatial correlation with exponential cutoff



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Estimation Methodology

The Clauset et al. (2009) Three-Step Recipe

1. **Parameter Estimation (MLE):**

$$\hat{\alpha} = 1 + n \left[\sum_{i=1}^{n} \ln \left(\frac{x_i}{x_{\min}} \right) \right]^{-1}$$

   For discrete data, solve transcendental equation with Hurwitz zeta function.

2. **Goodness-of-Fit Test:** Compute KS statistic between empirical and fitted CDF. Generate bootstrap samples from fitted power law. $p > 0.1$ indicates power law is plausible.

3. **Comparison with Alternatives:** Likelihood ratio test (Vuong's method) against log-normal, exponential, stretched exponential, and power law with exponential cutoff.

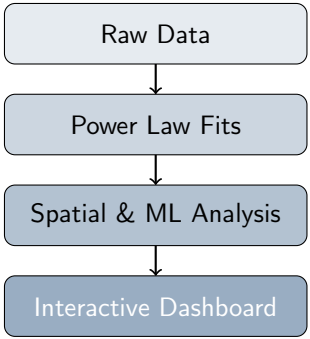
Key Requirement

Minimum sample size: $n \gtrsim 50$ for reasonable estimates, $n \gtrsim 1000$ in the tail for robust testing.

Data & Computational Framework

Data Sources

- **Infodengue API:** Weekly dengue case counts (2010–2025) for all 5,570 Brazilian municipalities
- **Episcanner:** Epidemic-level metrics (size, duration, peak week, R_0, onset, residuals)
- **Mosqlimate API:** Climate covariates (temperature, humidity) at quarterly resolution
- **Demographics:** Municipal population and socioeconomic indicators



Four-Pronged Analytical Approach

1. **Longitudinal α Tracking**
   - Yearly power law fitting for each city over 15 years
   - OLS regression on α trends to detect regime shifts
   - Identifies cities transitioning from predictable to extreme outbreak patterns

2. **Spatial Clustering (Local Moran's I)**
   - LISA analysis to identify hotspots of high/low α values
   - Detects geographic clusters of extreme outbreak propensity
   - Reveals spatial autocorrelation in scaling behavior

3. **Epidemiological Correlations**
   - α vs R_0, total cases, epidemic duration, onset timing, model residuals
   - Links statistical scaling to mechanistic epidemic drivers

4. **Predictive Modeling**
   - Random Forest, LightGBM (quantile), HistGradientBoosting (quantile)
   - Features: lagged metrics, α_{t-1}, $x_{\min}$, $\dot{\alpha}$, R_0, $\log_{10}(\text{Pop})$, climate
   - Targets: epidemic size S_t, duration D_t, peak week P_t



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Results: Spatial Distribution of α

Geographic Heterogeneity

- **Significant spatial variation** in scaling exponent across Brazilian states and municipalities
- States with **low** α (heavy tails): frequent extreme outbreaks, higher unpredictability?
- States with **high** α (thin tails): more predictable epidemic dynamics, fewer extreme events?

Interpretation of α

- Lower $\alpha \leftrightarrow$ thicker tail $\leftrightarrow$ higher probability of extreme outbreaks
- Analogous to Rhodes et al.'s finding that less transmissible diseases have **larger** τ (thinner tails)
- For dengue: α varies geographically, reflecting heterogeneity in transmission drivers (climate, demographics, immunity)

Key Insight

The spatial map of α serves as a **risk atlas**: regions with heavy-tailed dynamics require disproportionate preparedness capacity relative to their average case burden.

Results: Temporal Evolution of α

Longitudinal Scaling Trends

- Yearly α estimates reveal **temporal trajectories** of epidemic dynamics
- Some cities show **stable** α: consistent power law behavior over 15 years
- Others exhibit **regime shifts**: trending toward heavier or thinner tails

Trend Analysis

- OLS slope of α over time: $\dot{\alpha} = \frac{d\alpha}{dt}$
- $\dot{\alpha} < 0$: city becoming more prone to extreme outbreaks
- $\dot{\alpha} > 0$: city becoming more predictable (thinner tails)
- $\dot{\alpha}$ used as a predictive feature in the ML model

Connection to Komarova et al. (2020)

Temporal shifts in α may reflect the universal transition from exponential to power law growth as epidemics mature across spatially distributed populations.

Results: Epidemiological Correlations

Correlations with α:

- R_0**:** Basic reproduction number linked to tail thickness — higher transmission intensity associated with heavier tails
- **Total Cases:** α inversely related to attack rates — cities with lower α experience larger cumulative outbreaks
- **Epidemic Duration:** Heavy-tailed case distributions correlate with longer outbreak seasons
- **Onset Timing:** Earlier epidemic onset associated with different scaling regimes

Model Residuals:

- Sum of residuals from epidemic curve fitting vs α
- Cities with heavier tails show **larger residuals** in standard epidemic models
- This is the "**prediction gap**": conventional models systematically underperform in heavy-tailed regimes

Implication

The scaling parameter α is not merely a statistical artifact — it is an **emergent property** of the transmission system that encodes information about demography, climate, and epidemic dynamics.

Results: Epidemic Metrics Power Laws

Extended Power Law Analysis

Beyond weekly case counts, power laws also emerge in **epidemic-level metrics**:

- **Epidemic Size** (total cases per outbreak): heavy-tailed distribution confirmed via MLE + KS tests
- **Epidemic Duration** (weeks): also exhibits scaling behavior, though with different exponents

Combined CCDF Analysis

- Multi-year cumulative distribution functions with year-stratified visualization
- Yearly α trends overlaid on boxplots of metric distributions
- State-level aggregation reveals consistent patterns across aggregation levels

Consistency

Power law scaling in weekly incidence and epidemic size confirms that extreme outbreak dynamics are not limited to case counts — they are a **structural property** of the epidemic system.

Results: Predictive Modeling

Model Architecture

- Predict year t from year $t-1$ features
- **Models:**
  - Random Forest (baseline)
  - LightGBM (quantile loss)
  - HistGradientBoosting (quantile loss)
- **Targets:** S_t, D_t, P_t
- **Evaluation:** MAE, RMSE, Pinball Loss (quantile models), OOB R^2

Key Results

- Predicted vs Actual (log-log scale) shows good agreement across orders of magnitude
- Feature importance: lagged metrics + α features rank highly
- Quantile models (LightGBM, HGB) provide uncertainty bounds
- Population filter: removing small cities improves model performance

Residual Analysis

- Residuals vs Predicted: no systematic bias across prediction range
- Residual distributions approximately centered at zero
- Cities with >20% error analyzed by population distribution

Results: Size Classification Model — Predicting Large Epidemics

Objective

- Binary classification: will city i in year t exceed the p-th percentile of historical case counts?
- Target: $Y_{i,t} = \mathbb{I}\big(S_{i,t} > \text{percentile}_p(S_{\text{train}})\big)$, where the threshold is computed **only on training folds** to prevent leakage
- Default: $p = 75$-th percentile (configurable: 50–99)

Methodology

- **Models:** XGBoost, LightGBM classifiers
- **Validation:** TimeSeriesSplit (no shuffle) — preserves temporal ordering
- **Imbalance:** `scale_pos_weight` (XGBoost), `is_unbalance=True` (LightGBM)
- **Threshold optimization:** Youden's $J = \text{Sensitivity} + \text{Specificity} - 1$ on CV out-of-fold predictions

Features (X_{t-1})

- Lagged epidemic metrics: $S_{t-1}, D_{t-1}, P_{t-1}$
- Power law scaling: $\alpha_{t-1}, x_{\min,t-1}$
- Historical trend: $\dot{\alpha}_{t-1}$
- Epidemiological driver: $R_{0,t-1}$
- Demographics: $\log_{10}(\text{Pop})$
- Climate (quarterly): $\bar{T}_{Q1\ldots Q4}$, $\bar{H}_{Q1\ldots Q4}$

Key Findings and Implications

Main Results

1. **Power law scaling is pervasive** in dengue incidence across Brazilian municipalities — both in weekly case counts and epidemic-level metrics
2. **Spatial heterogeneity in** α reveals geographic regions at disproportionate risk of extreme outbreaks
3. **Temporal trends in** α identify cities undergoing regime shifts in epidemic dynamics
4. **Prediction gaps:** cities with heavy tails (low α) show reduced predictability in standard models
5. **Power law features improve forecasting:** incorporating α and its trend in ML models enhances prediction of extreme events
6. **Size Classification Model:** binary classifiers with temporal cross-validation can identify cities at risk of "Big Epidemics"



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Public Health Implications

Implications

- Risk assessment must account for **heavy-tailed outbreak distributions** — average predictions underestimate extreme event probability
- The α map provides a **prior for resource allocation**: heavy-tailed regions need disproportionate preparedness capacity
- Quantile regression models (LightGBM, HGB) offer **probabilistic forecasts** for worst-case scenarios
- The Size Classification Model enables **early warning systems**: flagging cities likely to exceed historical outbreak thresholds before the season begins, supporting proactive resource allocation and targeted interventions



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Acknowledgments & References

Acknowledgments

- Mosqlimate Group for data infrastructure and API access
- Brazilian Ministry of Health for Infodengue surveillance data
- Fundação Getulio Vargas (FGV) for computational resources
- The Wellcome Trust for funding

Software

- `powerlaw` (Python) — MLE fitting and comparison
- `PyGeoDa` — Local Moran's I clustering
- `scikit-learn`, `LightGBM` — Predictive models
- `Gradio` — Interactive dashboard

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Thank You!



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